Algebraic Notation and Substitution

Name _____ **Date** _____ No calculator Version 1

KEY WORDS variable · expression · term · substitute · formula

A Algebraic notation ●○○ — ANSWER ON THE SHEET

> **Remember:** In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $3 \times a$ = _____ **b)** $b \times 5$ = _____ **c)** $a \times b$ = _____ **d)** $a \times a$ = _____

e) $x \div 4$ = _____ **f)** $2 \times a \times b$ = _____ **g)** $y \times y \times y$ = _____ **h)** $1 \times n$ = _____

i) $7 \times m$ = _____ **j)** $p \div 2$ = _____ **k)** $n \times 6$ = _____ **l)** $3 \times a + 2 \times b$ = _____

B Writing expressions ●○○ — ANSWER ON THE SHEET

> ⚠ **Watch out:** "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 5 more than n _____ **b)** n multiplied by 4 _____

c) 6 less than n _____ **d)** n divided by 3 _____

e) double n _____ **f)** n squared _____

g) double n, then add 1 _____ **h)** subtract n from 10 _____

C Expressions for situations ●●○ — WORK IN YOUR BOOK

> **Remember:** Choose the operation that fits the story, then write it the short way. "3 dollars each for n pens" is $3 \times n$, written $3n$.

3. Write an expression for each situation.

a) The cost, in dollars, of n pens at \$3 each

b) The number of lollies left when you start with 20 and eat n of them

c) The age of Mia, who is 4 years older than her brother. Her brother is n years old.

d) The share each person gets when n lollies are shared equally between 5 people

e) The cost, in dollars, of a taxi that charges \$4 plus \$2 for each of n kilometres

f) The perimeter of a square with sides of n cm

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Substituting into an expression ●●○ — ANSWER ON THE SHEET

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

4. Find the value of each expression when $a = 3$ and $b = 5$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $10 - 2a$ = _____ **h)** $4b$ = _____

5. Find the value of each expression when $x = 6$ and $y = 2$.

a) $x - y$ = _____ **b)** $5y$ = _____ **c)** $\frac{x}{y}$ = _____ **d)** xy = _____

e) $x + 3y$ = _____ **f)** y^2 = _____ **g)** $2x - y$ = _____ **h)** $\frac{x}{2}$ = _____

E Substituting into a formula ●●● — WORK IN YOUR BOOK

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 8$ and $w = 5$.

b) The perimeter of a square is $P = 4l$. Find P when $l = 7$.

c) The cost in dollars of hiring a hall for n hours is $C = 5n + 20$. Find C when $n = 6$.

d) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 7$ and $w = 4$.

e) The distance a car travels in t hours is $d = 60t$. Find d when $t = 3$.

f) The number of matchsticks in pattern n is $T = 3n + 2$. Find T when $n = 10$.

★ Extension — FOR EARLY FINISHERS

7. Try different values of n to find the one that makes $3n + 1 = 16$ true.

8. Find the value of each expression when $a = 2$ and $b = 5$.

a) $a^2 + b^2$ **b)** $3(a + b)$ **c)** $\frac{4b}{a - 1}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

Algebraic Notation and Substitution

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS variable · expression · term · substitute · formula

A Algebraic notation ●○○ — ANSWER ON THE SHEET

> **Remember:** In algebra the × sign is left out and the number goes first: $a \times 3 = 3a$. Division is written as a fraction: $a \div 4 = \frac{a}{4}$.

1. Write each expression the short way.

a) $5 \times a$ = _____ **b)** $c \times 7$ = _____ **c)** $x \times y$ = _____ **d)** $b \times b$ = _____

e) $y \div 3$ = _____ **f)** $4 \times p \times q$ = _____ **g)** $a \times a \times a$ = _____ **h)** $1 \times m$ = _____

i) $9 \times k$ = _____ **j)** $t \div 5$ = _____ **k)** $n \times 8$ = _____ **l)** $2 \times x + 5 \times y$ = _____

B Writing expressions ●○○ — ANSWER ON THE SHEET

> ⚠ **Watch out:** "6 less than n" is $n - 6$, not $6 - n$. The order matters when you subtract or divide.

2. Write an expression for each statement.

a) 8 more than n _____ **b)** n multiplied by 6 _____

c) 4 less than n _____ **d)** n divided by 5 _____

e) triple n _____ **f)** n cubed _____

g) triple n, then subtract 2 _____ **h)** subtract n from 20 _____

C Expressions for situations ●●○ — WORK IN YOUR BOOK

> **Remember:** Choose the operation that fits the story, then write it the short way. "3 dollars each for n pens" is $3 \times n$, written $3n$.

3. Write an expression for each situation.

a) The cost, in dollars, of n books at \$8 each

b) The number of pages left when a book has 50 pages and you have read n of them

c) The age of Tom, who is 6 years older than his sister. His sister is n years old.

d) The share each person gets when n grapes are shared equally between 4 people

e) The cost, in dollars, of a taxi that charges \$5 plus \$3 for each of n kilometres

f) The perimeter of an equilateral triangle with sides of n cm

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.

D Substituting into an expression ●●○ — ANSWER ON THE SHEET

Remember: Replace each letter with its number, then calculate. If $a = 3$, then $2a = 2 \times 3 = 6$ (not 23).

4. Find the value of each expression when $a = 4$ and $b = 6$.

a) $a + b$ = _____ **b)** $2a$ = _____ **c)** ab = _____ **d)** $3b - a$ = _____

e) a^2 = _____ **f)** $2a + 3b$ = _____ **g)** $10 - 2a$ = _____ **h)** $4b$ = _____

5. Find the value of each expression when $x = 8$ and $y = 2$.

a) $x - y$ = _____ **b)** $5y$ = _____ **c)** $\frac{x}{y}$ = _____ **d)** xy = _____

e) $x + 3y$ = _____ **f)** y^2 = _____ **g)** $2x - y$ = _____ **h)** $\frac{x}{2}$ = _____

E Substituting into a formula ●●● — WORK IN YOUR BOOK

6. Use each formula.

a) The area of a rectangle is $A = lw$. Find A when $l = 9$ and $w = 6$.

b) The perimeter of a square is $P = 4l$. Find P when $l = 9$.

c) The cost in dollars of hiring a boat for n hours is $C = 8n + 15$. Find C when $n = 5$.

d) The perimeter of a rectangle is $P = 2l + 2w$. Find P when $l = 8$ and $w = 3$.

e) The distance a car travels in t hours is $d = 80t$. Find d when $t = 4$.

f) The number of matchsticks in pattern n is $T = 4n + 1$. Find T when $n = 10$.

★ Extension — FOR EARLY FINISHERS

7. Try different values of n to find the one that makes $4n - 3 = 17$ true.

8. Find the value of each expression when $a = 3$ and $b = 4$.

a) $a^2 + b^2$ **b)** $3(a + b)$ **c)** $\frac{4b}{a - 1}$

© 2026 Cell Tuition. You may print and photocopy this worksheet for your own classes and children. It may not be changed, sold or posted on other websites.